The Success of Tactical Combat Casualty Care as Seen Through Actor-Network Theory

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On my honor as a University student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments.

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Introduction

Although some loss of life is inevitable in combat, a large number of fatalities can be prevented, with approximately 25 percent deemed potentially survivable if treated properly (Eastridge et al., 2012). Of these preventable deaths, the majority are due to hemorrhage; data from 2001 to 2011 indicate that around 90 percent of avoidable fatalities in combat are due to uncontrolled bleeding (Eastridge et al., 2019; Peng, 2020). Previous analyses of battlefield trauma care have emphasized clinical outcomes and the importance of timely intervention (Blyth et al., 2015; Butler, 2017a).

In response to challenges observed during battlefield operations at the end of the 20th century, the U.S. military developed and implemented Tactical Combat Casualty Care (TCCC), a set of guidelines designed to reduce the number of preventable deaths due to hemorrhage (Butler & Kotwal, 2017). These guidelines were based on extensive review of trauma literature and battlefield reports, which highlighted gaps in extremity hemorrhage management (Blyth et al., 2015; Butler, 2017a). These guidelines succeeded primarily by restructuring battlefield trauma care and prioritizing hemorrhage control through technologies such as the tourniquet, and are now institutionalized across the U.S. military (Butler, 2017a). TCCC has been widely cited in medical and military literature as a model of evidence-based trauma care that integrates clinical innovations and scenario-based training (Butler, 2017a; Giebner, 2017).

TCCC is primarily evaluated through statistical outcomes, with an emphasis on the reduction of death rates and effectiveness of tourniquets. Within this literature, TCCC is portrayed as a successful medical innovation that improved survival in combat through effective guidelines and training. However, this focus on outcomes does not fully explain how TCCC was able to overcome resistance to tourniquets and reorganize battlefield trauma care into a coordinated system. Numerical success alone does not account for the sociotechnical process through which TCCC was implemented. Scholars have noted that understanding the effectiveness of complex systems such as TCCC requires attention to both human and non-human actors and procedures (Callaway, 2017; Cressman, 2009).

Without analyzing this transformation at the network level, TCCC appears to be a straightforward technological development rather than a reconfiguration of battlefield medical treatment. Such a limited understanding obscures the complexity of successfully introducing changes to a sociotechnical system. By focusing on clinical outcomes alone, existing studies overlook the organizational, procedural, and training structures that were critical to TCCC's adoption (Butler, 2017a; Giebner, 2017). This paper therefore asks: How did Tactical Combat Casualty Care become successfully integrated and institutionalized within the U.S. military despite prior resistance to key technologies such as the tourniquet?

This paper argues that Tactical Combat Casualty Care succeeded not merely because it introduced more effective treatment techniques and technologies, but also because its developers refocused hemorrhage as the central problem on the battlefield, redefined system roles, and instilled structured protocols, thereby stabilizing the trauma care system into a sociotechnical network (Butler, 2017a). Specifically, network builders translated diverse actors around a common goal of hemorrhage control, delegated medical decision-making to protocols, and stabilized the system through institutional mandates. To analyze this case, this paper uses Actor-Network Theory (ANT), which examines how networks of both human and non-human actors become stabilized. Using ANT concepts such as translation, delegation, and stabilization (Callon, 1987; Cressman, 2009), TCCC is interpreted not simply as a set of medical guidelines, but as a network-refining initiative that reconfigured battlefield trauma care. The perspectives summarized here are informed by prior scholarship documenting the clinical, organizational, and technological dimensions of TCCC implementation (Butler, 2017a; Callaway, 2017; Giebner, 2017). This argument is supported through analysis of policy, training, and empirical evidence. Together, these sources provide primary evidence of the construction and implementation of TCCC into a high-pressure, fast-paced environment.

Literature Review

Although several scholars have analyzed Tactical Combat Casualty Care (TCCC) and its impact on battlefield survival, existing literature focuses on clinical outcomes as a marker of success rather than sociotechnical challenges of implementing these. Most analyses simply treat TCCC as a medical innovation, failing to examine the network as a whole and how it achieved adoption and reached durability. As a result, there is currently a gap in the understanding of TCCC's success.

Butler provides a detailed account of TCCC in "Tactical Combat Casualty Care: Beginnings," describing the Naval Special Warfare research program that led to the creation of these guidelines for trauma care. This work emphasizes the integration of optimal medical practices with tactical awareness, highlighting innovations such as tourniquet use and scenario-based training. Although Butler effectively demonstrates the rationale and procedural improvements of TCCC, his analysis is limited by a failure to evaluate how these guidelines are applied under unpredictable battlefield conditions (Butler, 2017a).

In "The Transition to the Committee on tactical Combat Casualty Care," Giebner extends this discussion by examining the institutional and procedural mechanisms that ensured the success of TCCC through the foundation of the Committee on Tactical Combat casualty Care (CoTCCC). Unlike Butler, Giebner emphasizes the sociotechnical aspects of TCCC implementation, detailing the committee's iterative process. This process integrated perspectives from various sources including medics, corpsmen, surgeons, and civilian trauma specialists to ensure recommendations reflected both clinical evidence and battle constraints. Giebner's analysis highlights how structured decision-making and collaboration were crucial to the translation of evidence-based protocols into actionable guidelines (Giebner, 2017).

Together, Butler and Giebner provide complimentary insights into the implementation of TCCC with Butler focusing on the clinical innovations and Giebner describing the organizational, procedural, and human factors involved. However, a gap remains regarding how these clinical and organizational elements functioned together, forming a stabilized network. Existing analyses fail to investigate how actors within the system navigate logistical constraints, tactical pressures, and situational variability when applying these guidelines.

This paper addresses that gap by exploring the intersection of clinical protocol, human factors, and situational context. It argues that the success of TCCC depends not only on procedures, but also on the decision-making structures and training strategies that enable their consistent and effective use in combat settings. By examining this, the study advances the understanding of how to optimize both the design and implementation of battlefield trauma care, addressing limitations in prior scholarship that focus solely on clinical efficacy.

Conceptual Framework

This paper analyzes the development and implementation of Tactical Combat Casualty Care (TCCC) using Actor-Network Theory (ANT), a Science and technology Studies (STS) framework developed by Michel Callon, Bruno Latour and John Law to explain how scientific and technological systems emerge through networks of human and non-human actors rather than through isolated technical innovation, emphasizing that success depends on the alignment and stabilization of heterogeneous actors within a sociotechnical system (Cressman, 2009). ANT is relevant to this case as it allows for analysis of TCCC not simply as a medical innovation or set of guidelines, but as a sociotechnical network of human and non-human actors whose relationships require stabilization. Rather than treating technology as a neutral tool inserted into a preexisting system, ANT examines how different technologies, people, and institutions are arranged to form a functional network. This framework therefore makes it possible to explain the institutionalization of TCCC across the U.S. military due to successful reorganization of battlefield trauma care into a stabilized network. Key ANT concepts that guide this analysis include actors, translation, delegation, and stabilization, each of which explains how sociotechnical systems are constructed and maintained over time. In ANT, sociotechnical networks are formed by network builders: individuals or groups who actively assemble human and non-human actors into a coordinated system to accomplish a specific goal (Cressman, 2009). In the case of TCCC, the network builders are the military physicians, surgeons, and combat medical leaders who developed and promoted TCCC guidelines.

The central premise of ANT is that both human and non-human elements participate in shaping outcomes together. ANT refers to these elements as actors, emphasizing that agency is distributed across a network rather than with specific individuals (Cressman, 2009). In this case, actors may include medics, commanding officers, training protocols, tourniquets, clinical data, battlefield conditions, written guidelines, and many other entities that interact to form the battlefield care system. The network builders coordinate these actors and work to align them around the shared goal of improving battlefield survival through effective hemorrhage control and standardized trauma care.

ANT conceptualizes the formation of sociotechnical networks primarily through the process of translation, which is considered the central mechanism through which actors become aligned. Rather than assuming a separation between social and technical aspects, translation describes the creation of relationships between various elements, the combination of which is a key factor in analyzing networks. Within ANT, translation bridges heterogeneous components by creating overlaps between them, establishing relationships. Through translation, actors come to share aligned goals, forming a sociotechnical network (Cressman, 2009). Through translation, network builders align actors by defining problems, assigning roles, and creating shared objectives, guiding the development of the network.

While translation explains how network elements are connected and aligned, ANT also emphasizes how networks become stable over time through delegation: a specific form of translation where responsibilities are shifted from humans to technological innovation, with technologies shaping and constraining behavior, therefore structuring action within a network. Delegation thereby demonstrates how societal factors become embedded in technologies, particularly those that organize human actions (Cressman, 2009). Network builders use delegation to embed decision-making into structured protocols and technologies, ensuring that battlefield care is performed consistently.

Closely related to delegation is stabilization, the process of a sociotechnical network becoming stable over time. A network is considered stabilized when its actors reliably perform their defined roles and delegated responsibilities are consistently completed through established protocols. Stabilization refers to the sustained durability of a network as a result of successful alignment and delegation (Jarrahi & Sawyer, 2019). Drawing on these concepts from ANT, the following analysis will first examine the processes of translation that align actors, followed by how responsibilities are delegated, and finally how these processes contribute to the stabilization of a sociotechnical system. These concepts allow ANT to explain not only how TCCC was developed, but how it became widely accepted and institutionalized. Throughout the following analysis, the term network builders will be used to refer to the actors responsible for forming and maintaining the TCCC network and ensuring its long-term institutionalization. The following analysis uses translation to examine hemorrhage control as the central problem, explains how delegation through structured protocols made medical decisions consistent, and shows how stabilization ensured TCCC's long-term institutionalization following its introduction.

Analysis

Using Actor-Network Theory, this analysis interprets the implementation of Tactical Combat Casualty Care (TCCC) as the construction of a sociotechnical network. Drawing on ANT, this analysis examines how network builders aligned actors through translation, delegation, and, ultimately, stabilization of the battlefield trauma care system. Together, these concepts provide insight into why TCCC succeeded not merely as a set of clinical innovations, but as a durable network of actors.

Translation

The initial success of TCCC depended on the effective translation of actors. This was achieved by aligning diverse participants around the shared goal of controlling hemorrhage on the battlefield. Translation allowed the network builders to connect medics, commanding officers, technology providers, and others by aligning their interests toward a common problem: preventable battlefield deaths due to hemorrhage.

A critical early step in this translation process was the establishment of the Naval Special Warfare (NSW) Biomedical Research Program in 1989, a program tasked with addressing medical issues relevant to Navy SEAL operations, including battlefield trauma care. This program identified extremity hemorrhage as the leading cause of preventable death in combat and highlighted gaps in military medical practices such as the lack of tourniquet use. To address this, the TCCC Research Project was launched (Butler, 2017a). By producing evidence-based recommendations, the NSW program helped align diverse actors around a shared understanding of the need to reduce preventable deaths. By defining hemorrhage as the primary problem, TCCC reframed battlefield medicine around a specific cause of death. This reframing represents translation in ANT: heterogeneous actors were aligned around a newly prioritized problem.

For example, Chabel's research on tourniquet-induced limb ischemia in 1990 (Chabel et al., 1990) documented the physiological risks associated with tourniquet use. During the early 1990s, there was a strong stigma against tourniquets, with many, including medical personnel, believing them to be unsafe and cause ischemia, which is tissue damage or death (Bober & Jahangir, 2015). Debates over tourniquet use illustrate the power of translation. Despite data from the Vietnam War indicating hemorrhage to be the leading cause of death, with 87 percent of deaths in the first 24 hours due to uncontrolled bleeding (Blyth et al., 2015), it was widely believed that tourniquets should be used as a last resort for fear of ischemic injury.

The introduction of this data from Vietnam functions as evidence, demonstrating a clear mismatch between cause of death and treatment hesitation. The significance lies in revealing a misalignment within the existing network: clinical caution was prioritized over statistical lethality. The network builders within the NSW Biomedical Research Program and subsequent TCCC workshops provided both medical evidence and tactical reasoning to accomplish this, aligning actors around safe tourniquet use in battlefield care (Butler, 2017a). By connecting scientific findings with operational constraints, these efforts resolved initial resistance and demonstrated how heterogeneous actors could converge on a shared practice.

Therefore, translation did not simply spread information; it created a network of actors who could operate in a coordinated way. The NSW Biomedical Research Program produced the evidence base by generating data on extremity hemorrhage and tourniquet use that challenged beliefs at the time. This data was then translated into action on the battlefield through scenario-based training (Butler, 2017a). This illustrates how translation aligns both social and technical actors, bridging previously separate domains to form the foundation of a reliable battlefield trauma care system. Translation allowed diverse groups to converge on hemorrhage control as the central goal, enabling the coordination of the TCCC network. Without this alignment, widespread adoption of tourniquet use and other guidelines would be impossible.

Delegation

TCCC delegated key responsibilities from human actors to structured protocols, guidelines, and technologies. This allowed the system to function reliably under different combat pressures and scenarios. Through delegation, technologies and protocols perform work that would otherwise require continuous human attention (Cressman, 2009).

The 1996 TCCC Guidelines listed by Butler in "Tactical Combat Casualty Care: Beginnings" provide a clear example of how delegation structured battlefield medical care. Care was divided into three phases: Care Under Fire, Tactical Field Care, and Casualty Evacuation Care. These guided medics on when and how to act in unpredictable, complex situations on the battlefield. Tourniquet use was strongly recommended for extremity hemorrhage, removing the need for medics to debate its use. Scenario-based training further delegated decision-making by turning complex battlefield situations into practiced responses. This enabled medics to act automatically and consistently in a variety of scenarios. Additional recommendations further reduced the cognitive load experienced by human actors by embedding evidence-based medical decisions directly into the guidelines (Butler, 2017a).

Network builders designed scenario-based TCCC training and structured protocols that shifted battlefield decision-making from spontaneous judgments to automated, reliable responses. The first published report of success with TCCC in the Army Medical Department Journal was by an Army surgeon who stated that "Tourniquets played a decisive role in quickly and effectively stopping hemorrhage under fire and keeping a number of Soldiers with serious extremity wounds involving arterial bleeding alive until they could eventually undergo emergent surgery at the Forward Surgical Team" while at war in Baghdad (Butler, 2017b). This quotation serves as primary evidence of delegated protocol functioning under combat conditions. This is significant as it demonstrates that medics were not improvising, they were enacting pre-structured decisions embedded in TCCC guidelines. This example shows that medics, guided by TCCC protocols and training, were able to act swiftly and correctly under high-stress conditions following the introduction of TCCC, reducing reliance on decision-making in the moment.

Furthermore, from 1996 when the initial TCCC guidelines were released, to 2011, the number of combat fatalities due to extremity hemorrhage decreased by 67 percent from 7.8 to 2.6 percent of total deaths (Butler, 2017b). This statistic functions as quantitative evidence that delegated practices were enacted consistently across units, proving that delegation produced repeatable outcomes, not isolated success. This highlights how network builders used TCCC protocols to effectively delegate critical medical decisions to structured guidelines, ensuring medics could act confidently in high-pressure battlefield scenarios.

Delegation, therefore, illustrates the interdependency of social and technical elements of this network as human action became embedded in procedures, structuring medical behavior as a result. By reducing reliance on individual medics' decision-making, TCCC protocols ensured consistent, effective care and allowed medics to focus on execution rather than deliberation. Without this delegation, the variability of human judgment could have undermined the stability and effectiveness of TCCC.

Counterargument

The previous analysis demonstrated that TCCC's success relied on translation and delegation to align actors and structure battlefield medical care. However, some might argue that the program's effectiveness was primarily the result of technological improvements rather than coordination of the system. From this perspective, devices such as tourniquets and dressings alone led to the reduction in preventable deaths seen following TCCC, making the coordination by network builders of actors within the network a primary concern, rather than a secondary one.

However, this interpretation overlooks the critical role of sociotechnical alignment in enabling consistent, effective care. The success following the development of the Committee on Tactical Combat Casualty Care (CoTCCC), emphasizes that the reduction in preventable deaths due to hemorrhage is a result of the systematic application of structured protocols and coordinated training, rather than the presence of medical devices. Network builders within this committee were responsible for maintaining and updating the TCCC guidelines based on new technologies and information, but primarily introduced systemic, function-based changes to the guidelines in the 2000s. Following its creation, preventable deaths in combat continued to decrease, with CoTCCC commonly being credited (Callaway, 2017). This new evidence responds directly to the claim that improved technology is responsible for the reduction in hemorrhage fatalities by showing that organizational restructuring and iterative review, correlated with sustained improvements. If technology alone explained success, ongoing committee intervention would be unnecessary.

Thus, while technological innovations were necessary for the reduction of preventable deaths in combat, ANT highlights that TCCC's success emerged from the organization of a network containing human actors, structured procedures, and devices that must interact effectively. The improvement of combat trauma care following the development of TCCC cannot be explained by technology alone. It depends on translation and delegation to ensure reliable application in the field.

Stabilization

Stabilization of TCCC occurred as network builders ensured that actors within the network consistently adhered to their delegated roles, with the guidelines becoming embedded within the military medical system. In Actor-Network Theory, stabilization reflects the moment when a network achieves durability. This is characterized by actors sharing aligned goals, procedures becoming standardized, and practices being enacted consistently across varying contexts (Jarrahi & Sawyer, 2019). In the case of TCCC, stabilization is evident due to its long-standing institutionalization across the Department of Defense.

As of 2018, TCCC is the official standard for prehospital combat casualty care across the U.S. military. Department of Defense Instruction (DoDI) 1322.24, "Medical Readiness Training," formally required all service members to maintain current TCCC certification (Gurney et al., 2020). This required both initial and recurring training for all military first responders, thereby embedding TCCC across the military rather than allowing individual units discretion. This policy functions as an actor within the ANT framework, enforcing alignment and making participation in the TCCC network obligatory rather than voluntary.

Network builders implemented this formal mandate, representing structural stabilization at an institutional level. The DoDI does not introduce new medical technology but rather reinforces compliance with an existing set of guidelines. Essentially, this strengthened TCCC alignment across actors within the network by applying it across the full system of military medicine. The durability, and therefore long-term success, of TCCC is therefore proven not to be due to specific technologies, but its application.

Survey data further demonstrates how stabilization depends on consistent network engagement. Those who believed that 80 to 100 percent of their medical team had completed TCCC training had significantly higher confidence in their medics' abilities on the battlefield than those who believed none of their staff was trained with Adjusted Odds Ratios (AOR) of 2.34 and 0.28 respectively (Gurney et al., 2020). The survey data serves as evidence that institutional mandates translate into behavioral confidence differences, meaning that stabilization is not merely policy-based, but enacted through collective compliance

While the DoDI demonstrates stabilization of TCCC through formal policy, the survey findings provide evidence of it in practice. Confidence levels varied significantly depending on perceived TCCC training completion rates among medical personnel, indicating that durability depends on the collective application of guidelines, rather than their existence. A network remains strong only when the actors within it consistently adhere to their delegated roles. This demonstrates that stabilization is both behavioral and institutional as policies must be translated into routine practice across all units to achieve durability. Therefore, the stabilization of TCCC as a network indicates that its success is not solely due to available technologies, but also due to sustained coordination of actors performing in alignment.

Analysis Conclusion

Evaluating TCCC through Actor-Network Theory demonstrates that its success cannot be explained by technological innovation alone. Through translation, military leaders and medical experts aligned diverse perspectives around a shared goal; through delegation, structured protocols embedded evidence-based decision-making into battlefield practice; and through stabilization, these practices were institutionalized across the U.S. military. The reduction in preventable deaths following the creation of TCCC can thereby be accredited to the coordinated interaction of different actors including medical staff, military leaders, policies, training systems, and medical devices within its network. TCCC therefore illustrates how durable organizational change in high-risk environments depends on sociotechnical alignment rather than technological advancement alone.

Conclusion

This paper demonstrates that the success of Tactical Combat Casualty Care (TCCC) cannot be attributed solely to technological innovations or medical devices such as the tourniquet. Using Actor-Network Theory, it has been shown that TCCC succeeded through the alignment of actors through translation, the delegation of critical decisions to structured protocols, and the stabilization of these practices across the U.S. military by network builders. Together, these transformed battlefield trauma care into a durable sociotechnical network, allowing for consistent, effective hemorrhage control and thereby the reduction preventable combat deaths.

The significance of this analysis lies in its reframing of TCCC as more than a set of clinical guidelines. By highlighting the sociotechnical processes that enabled its success, the study provides a deeper understanding of how organizational, human, and technological elements interact within high-pressure networks. This insight can aid as a basis for future research on implementing complex systems in similarly challenging environments, emphasizing that both social and technical factors must be aligned for success.

In engineering, these findings suggest that successful system design depends not only on the relevant tools, but also on their integration within a structured network of protocols, training, and human actors. This means that the context and interactions surrounding a tool are just as important as the tool itself. Engineers should therefore consider the sociotechnical network throughout the design process, ensuring that innovations are not only effective in isolation, but also reliably integrated into practice.

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